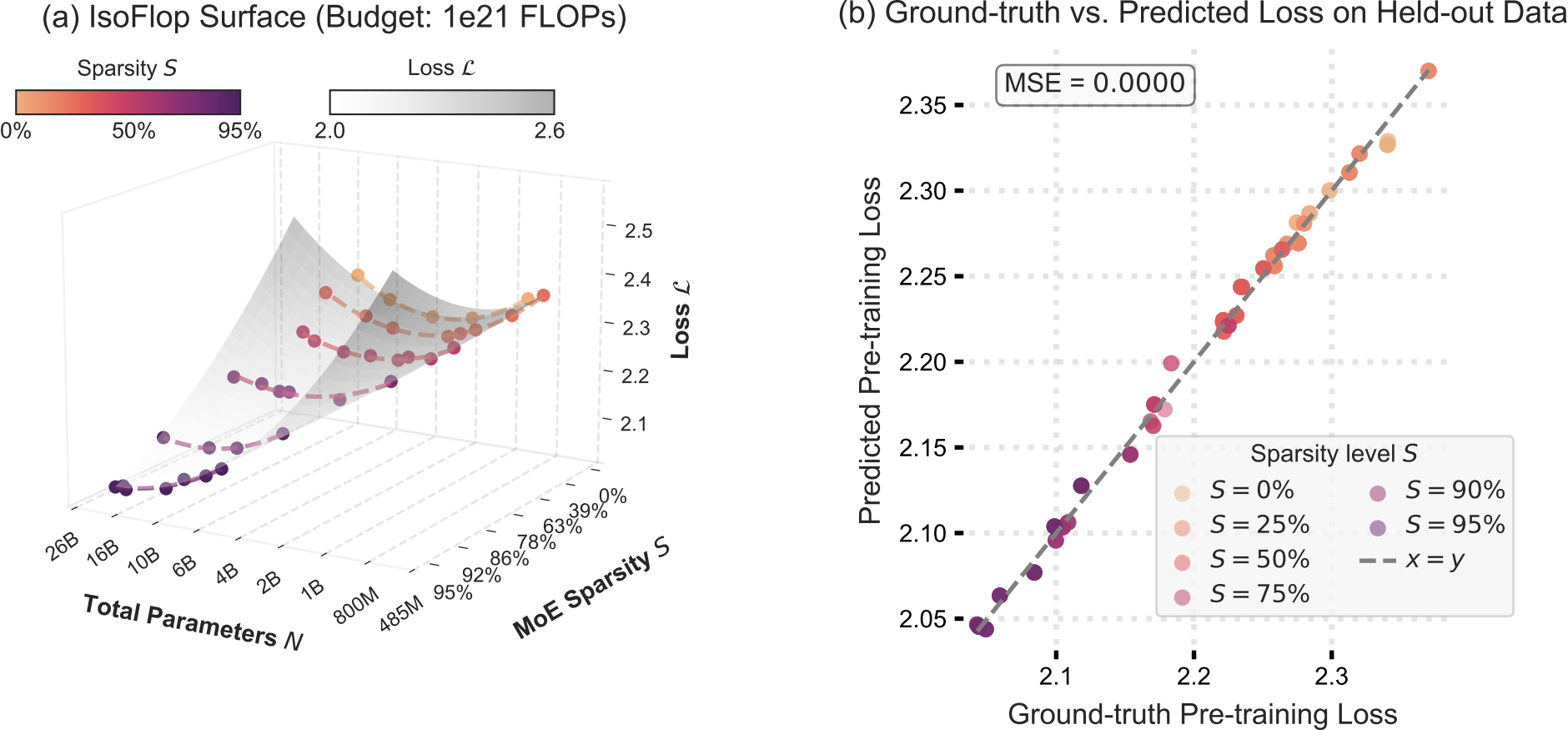
(a) IsoFlop Surface (Budget: 1e21 FLOPs)
Sparsity S
0%
50%
95%
Loss $\mathcal{L}$
2.0
2.6
2.5
2.4
2.3
2.2
2.1
Loss $\mathcal{L}$
26B
16B
10B
6B
4B
2B
1B
800M
485M
Total Parameters N
0%
39%
63%
78%
86%
92%
95%
MoE Sparsity S
(b) Ground-truth vs. Predicted Loss on Held-out Data
MSE = 0.0000
Predicted Pre-training Loss
2.35
2.30
2.25
2.20
2.15
2.10
2.05
2.1
2.2
2.3
Ground-truth Pre-training Loss
Sparsity level S
$S = 0\%$
$S = 25\%$
$S = 50\%$
$S = 75\%$
$S = 90\%$
$S = 95\%$
$x = y$